

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester B. Tech. (ME) Examination****September 2018****ME 245 Manufacturing Processes - II****Date: 17.09.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Explain the following lathe operations. (each carry one mark) [07]**

- (i) Facing
- (ii) Grooving
- (iii) Knurling
- (iv) Boring
- (v) Parting off
- (vi) Drilling
- (vii) Metal spinning

Q - 2. Answer any three questions. [12]

- (i) Explain the steady rest and follower rest as used in engine lathes. Give their function.
- (ii) Differentiate between Engine lathe and capstan and turret lathe.
- (iii) List main parts of engine lathe and state functions of each part.
- (iv) Explain the construction & working of Upright drilling machine with neat sketch.
- (v) Explain with neat sketch :
 - a) Counter-Boring
 - b) Counter-Sinking

Q - 3 Answer any four questions. [16]

- (i) Calculate the crank movement for 87 divisions with the help of compound indexing method.
- (ii) Sketch & compare Climb milling and Conventional milling.
- (iii) Explain the crank and slotted lever mechanism for the shaper machine.
- (iv) Differentiate between Shaper and Planner with reference to its construction, operation and applications.
- (v) Explain the operations of centre less grinding and state its advantages.
- (vi) Explain the basic principle of Broaching. How it differs from shaping operation?

SECTION – II

Q - 4 Answer the following questions.

- (i) What do you mean by sheet, plate and slab? [03]
- (ii) List the advantages of hot working process over cold working process. [04]
- (iii) Write a short note on residual stress. [04]

Q - 5 Answer any two questions.

[12]

- (i) Explain any two rolling mills with neat sketch.
- (ii) Prove equation for maximum draft $(\Delta h)_{\max} = \mu^2 R$ where μ is coefficient of friction and R is roller radius.
- (iii) Explain forging defects.
- (iv) What do you understand by the term flash in a forging? Explain with the help of a sketch. By which operation flash can be removed?

Q - 6 Answer any two questions.

[12]

- (i) Explain cold extrusion forging process with neat sketch.
- (ii) How the wire drawing process is done? Explain with the help of sketch.
- (iii) Explain embossing and coining processes with neat sketches.
- (iv) Determine the die & punch sizes for blanking a circular disc of 20 mm diameter from a C20 steel sheet having thickness 1.5 mm. Shear strength of the material is 294MPa.
